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Studierichting: Informatica

Oefeningenreeks 6A nr. 25.3

De tweede verschillen zijn constant, dus de rij is van orde 2.

$$\Rightarrow x_n = an^2 + bn + c$$

$$\begin{cases} a + b + c = 4 \\ 4a + 2b + c = 6 \\ 9a + 3b + c = 6 \end{cases}$$
$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 4 & 2 & 1 & 6 \\ 9 & 3 & 1 & 6 \end{array} \right]$$

$$R_2 \leftarrow R_2 - 4R_1, R_3 \leftarrow R_3 - 9R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -2 & -3 & -10 \\ 0 & -6 & -8 & -30 \end{array} \right]$$

$$R_3 \leftarrow R_3 - 3R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -2 & -3 & -10 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$\Rightarrow \boxed{c = 0}$$

$$-2b - 3c = -10$$

$$\Rightarrow \boxed{b = 5}$$

$$a + b + c = 4$$

$$\Rightarrow \boxed{a = -1}$$

$$\Rightarrow x_n = -n^2 + 5n$$

$$x_8 = -8^2 + 5 \cdot 8 = -64 + 40 = -24$$

References